

DRONEARJUNA (DA) — PROJECT PLAN

Closing the Phase 3–6 Gaps

Ten weeks, two staffing windows, one fleet-management build.

PLAN PERIOD	BASELINE
10 August 2026 – 30 September 2026	DA-Implementation-Status.pdf, 5 Aug 2026
TEAM	CLASSIFICATION
Chandra Sekhar (Backend Lead) · Vaishnavi (Backend) · Indra (Frontend)	Restricted / Confidential

01 — PURPOSE & STAFFING WINDOWS

Two windows, one constraint

This plan sequences every gap identified in the As-Built Implementation Status Report (Phases 3–6) across two staffing windows, built around a single constraint: Vaishnavi's availability drops to a quarter of normal for the first three weeks.

Window A — 10–31 Aug. Vaishnavi at 25% capacity. Chandra Sekhar carries the backend load; Indra runs frontend independently. · **Window B — 1–30 Sep.** Vaishnavi returns to full capacity — backend work splits in parallel between her and Chandra Sekhar, while Indra continues frontend.

Effort below is costed in **person-days (PD)** — one PD is one person at 100% capacity for one day. Because Vaishnavi works at 25% through Window A, her elapsed calendar time to deliver a task runs roughly four times its PD estimate.

Fourteen gaps, costed

Every item from the status report's gap summary, assigned a track and an effort estimate. Dependencies determine the sequencing in Section 4.

Ref	Gap	Track	PD	Depends on
G1	3D visualisation scaffold (Cesium/terrain, 3D drone tracking)	FRONTEND	10	none
G2	Telemetry replay player (3D + timeline scrub)	FRONTEND	5	G1
G3	Drone Inventory — full knowledge-base workflow (linked records, cross-refs, browse UI)	BOTH	8	none
G4	Drone Analyst — background worker infrastructure (job queue consumer)	BACKEND	6	none
G5	Drone Analyst — object-detection inference (YOLOv8 integration)	BACKEND	8	G4
G6	Drone Analyst — video-frame extraction pipeline	BACKEND	5	G4
G7	Drone Analyst — change-detection module	BACKEND	6	G5, G6
G8	Object storage wired into code (MinIO upload: imagery/video/detections)	BACKEND	4	none
G9	Drone Analyst frontend workspace (job launch, results viewer, dashboard)	FRONTEND	8	G4, G5
G10	Automated test coverage — raise to ≥80%, verified	BOTH	6	ongoing
G11	CI pipeline (lint, type-check, test, build on PR)	BOTH	4	none
G12	Accessibility pass (WCAG across all 5 workspaces)	FRONTEND	5	none
G13	Load testing (telemetry pipeline + API under multi-drone load)	BACKEND	4	G4–G8 stable
G14	Frontend automated test suite (currently zero frontend tests)	FRONTEND	6	none

03 — CAPACITY MODEL

What each window can actually absorb

Person	Window A (16 wd)	Window B (22 wd)
Chandra Sekhar	16 PD (100%)	22 PD (100%)
Vaishnavi	4 PD (25%)	22 PD (100%)
Indra	16 PD (100%)	22 PD (100%)

Backend capacity: **20 PD** in Window A, rising to **44 PD** in Window B. Frontend capacity is Indra alone throughout: **16 PD**, then **22 PD**.

04 — SEQUENCING LOGIC

Why the order is the order

Window A backend — 19 PD against 20 PD capacity

Ref	Priority item	PD
G4	Analyst background worker infrastructure	6
G8	Object storage wiring	4
G3	Inventory knowledge-base data/link model (backend half)	5
G10	Coverage tooling + first gap-fill pass (start)	4

Window A frontend — 16 PD against 16 PD capacity

Ref	Priority item	PD
G1	3D visualisation scaffold	10
G14	Frontend test suite scaffolding + first pass (start)	6

Window B backend — ~25–29 PD against 44 PD capacity

Ref	Priority item	PD
G5	Object-detection inference (YOLOv8)	8
G6	Video-frame extraction pipeline	5
G7	Change-detection module	6
G10	Coverage to $\geq 80\%$, verified (finish)	—
G13	Load testing	4
G11	CI pipeline — backend jobs	2

Window B frontend — ~23 PD against 22 PD capacity

Ref	Priority item	PD
G2	Telemetry replay player	5
G9	Drone Analyst frontend workspace	8
G12	Accessibility pass	5
G14	Frontend test suite (finish)	—
G3	Inventory knowledge-base browse UI (frontend half)	3
G11	CI pipeline — frontend jobs	2

05 — INDIVIDUAL TASKS • WINDOW A

10 August – 1 September

Chandra Sekhar

Backend Lead

100% • 16 PD

Aug 10–13	Design + implement Analyst job-queue consumer — RabbitMQ worker process, job state transitions queued → running → done/failed.	G4
Aug 14–17	Wire MinIO object storage into the Analyst job pipeline — upload source imagery, store object keys against job records.	G8
Aug 18–21	Inventory knowledge-base data model — link table between drones/threats/payloads, cross-reference queries, backend endpoints.	G3
Aug 24–28	Backend coverage tooling (pytest-cov + reporting) — baseline report, begin closing gaps in weakest modules.	G10
Aug 31	Integrate Vaishnavi's G4 test/doc output; buffer & catch-up day.	—

Vaishnavi

Backend

25% • 4 PD

Aug 10–14	Job-registry schema review; unit tests for job state-transition logic, paired with Chandra Sekhar's G4 consumer. (1.5 PD)	G4
Aug 17–21	Integration tests for the MinIO upload path — mock bucket and real bucket cases. (1.5 PD)	G8
Aug 24–31	Document Analyst-worker and object-storage design decisions as a DA-DDD-001 addendum. (1.0 PD)	G4/G8

Fractional time is assigned to test-writing and documentation on Chandra Sekhar's in-progress work rather than an independent task — this avoids context-switch overhead a standalone task would carry at 25% availability, and keeps her productive without becoming a blocking dependency for anyone.

Indra

Frontend

100% • 16 PD

Aug 10–14	Evaluate and scaffold Cesium integration into the Fly workspace — terrain tile loading, camera controls.	G1
Aug 17–21	3D live drone tracking — render live telemetry positions as 3D entities on the globe, synced with existing 2D map state.	G1
Aug 24–26	3D visualisation polish — altitude-exaggeration toggle, drone model/icon, mission waypoint overlay in 3D.	G1
Aug 27–31	Frontend test suite scaffolding (Vitest + React Testing Library); first tests for Fleet workspace and shared components.	G14

06 — INDIVIDUAL TASKS • WINDOW B

1 September – 30 September

Chandra Sekhar

Backend Lead

100% • 22 PD

Sep 1–4	YOLOv8 model integration — load pretrained weights, inference endpoint wired to the job worker.	G5
Sep 7–9	Inference results → detection-result table persistence; finalise confidence/bounding-box schema.	G5
Sep 10–14	Video-frame extraction pipeline — ffmpeg-based frame sampling from uploaded video, feeding the inference queue.	G6
Sep 15–17	CI pipeline, backend jobs — lint, type-check (mypy/ruff), pytest run, Docker build in GitHub Actions.	G11
Sep 18–21	Load testing — simulate 10+ drone telemetry load, profile and fix the bottlenecks found.	G13
Sep 22–25	Change-detection module — pairwise frame/image comparison logic, threshold tuning.	G7
Sep 28–30	Final coverage gap-fill push to ≥80%; review Vaishnavi's parallel output, integration pass.	G10

Vaishnavi

Backend

100% • 22 PD

Sep 1–4	Inventory knowledge-base remainder — full-text cross-reference search across drones/threats/payloads, admin link-management endpoints.	G3
Sep 7–10	Change-detection supporting utilities — image diff scoring, before/after pair storage, run in parallel with Chandra Sekhar's core logic.	G7
Sep 11–14	Backend coverage gap-fill — target drone_analyst, drone_inventory, mavlink_broadcaster edge cases.	G10
Sep 15–18	Object-storage lifecycle — retention/cleanup policy for old job artifacts, presigned URLs for frontend result viewing.	G8

Sep 21–24	Load-test harness and fixtures for simulated multi-drone load, supporting Chandra Sekhar's profiling run.	G13
Sep 25–30	Analyst API polish — pagination on job/results list endpoints; final integration testing with Indra's Analyst UI.	G10

Runs in parallel with Chandra Sekhar throughout — Window B is the first point at which backend has two full-time engineers, so the two tracks are deliberately independent except where explicitly noted as paired (G7, G13).

Indra Frontend		100% · 22 PD
Sep 1–4	Telemetry replay player — timeline scrub UI, playback controls, synced with the 3D view from G1.	G2
Sep 7–10	Analyst workspace — job launch UI: upload/select source, launch job, status polling.	G9
Sep 11–14	Analyst workspace — results viewer: detection overlays, model registry browser, telemetry stats dashboard.	G9
Sep 15–17	CI pipeline, frontend jobs — lint, type-check, Vitest run, build in GitHub Actions.	G11
Sep 18–22	Accessibility pass across all 5 workspaces — keyboard nav, ARIA labels, contrast, screen-reader pass.	G12
Sep 23–25	Inventory knowledge-base browse UI, consuming Vaishnavi's Sep 1–4 backend work.	G3
Sep 28–30	Frontend test suite — finish coverage on Fly/Plan/Monitor workspaces.	G14

07 — MILESTONES

Checkpoints across the ten weeks

21 Aug	Analyst job worker + object storage live end-to-end (backend).
31 Aug	3D visualisation scaffold usable in the Fly workspace; Window A close-out review.
04 Sep	YOLOv8 inference producing real detection results on test imagery.
14 Sep	Change-detection and video pipeline functional; Analyst workspace UI usable end-to-end.
21 Sep	CI pipeline green on both backend and frontend; load-test report delivered.
30 Sep	≥80% verified coverage, accessibility pass complete, all Phase 3–5 gaps closed or explicitly deferred.

08 — RISKS & NOTES

What could move this plan

YOLOv8 integration (G5) is the highest-uncertainty item. Model performance on drone-specific imagery is unproven — if accuracy is poor, budget an extra 3–5 PD in late September for retraining, drawn from the G13 load-testing buffer if needed.

Vaishnavi's Window A tasks are deliberately paired with Chandra Sekhar's, not independent — 25% availability spread over three weeks makes standalone ownership of anything with an external dependency high-risk for slippage.

3D visualisation (G1) has no backend dependency and is scheduled first in Window A so Indra is never blocked waiting on backend work.

The CI pipeline (G11) is intentionally mid-plan, not first — sequencing it once G1 and G4 exist gives it real code paths to lint, test, and build against, rather than standing it up against an incomplete tree.